Adaptive CMF multiplication orders and a stronger arithmetic divisor

An addendum for the direction $(10, 9, 11, -11; 2, 1, 3)$

Exact path experiments and a uniform arithmetic extension

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Abstract

We test literal changes of native CMF path, independently for each prime and each prescribed initial row. Prime-dependent path selection strictly improves finite divisibility certificates. Different rows can have incompatible maximizing paths and a strictly better path-only divisor; after adjoining the previous arithmetic normalization, separate row choices give no further common-divisor gain in the tested families. The residual prime pattern leads to a uniform extension of the previous six-interval arithmetic lemma to all integer translates of its carry chambers. Combined with the previously established analytic estimates, the extension gives $\mu(\zeta(2)) \le 62636575/13646619 < 4.589897$. The path experiments and the uniform readout argument are distinguished throughout: the new measure bound is not attributed to an unproved row-dependent ordering rule.

1 What was tested, and what improved

The CMF is the native pFq$(4, 3, 1)$ field, with

$$w_0 = (1, 1, 1, 0; 1, 1, 1), \quad v = (10, 9, 11, -11; 2, 1, 3), \quad R = \begin{pmatrix} 0 & 2 & -1 \\ 2 & 0 & 0 \end{pmatrix}.$$

Write $W_N = W(w_0, w_0 + Nv)$ and $(p_N^{\mathrm{raw}}, q_N^{\mathrm{raw}})^T = RW_N e_0$. Conservation identifies admissible paths with the same endpoint. It does not permit unshifted matrices to be commuted.

For each cut vertex w, compute the complete native products $A_w = W(w_0, w)$ and $B_w = W(w, w_0 + Nv)$, and verify all nine entries of $A_w B_w = W_N$ exactly. At this cut the certificate for initial row r and prime p is

$$b_{p,r}(w) = \min_{0 \le i \le 2} v_p\big((RA_w)_{r,i}(B_w)_{i,0}\big).$$

It bounds the valuation of the endpoint scalar, since it bounds every term in the three-term sum. Entire blocks are evaluated before taking this minimum. The suffix always retains at least $2N$ negative-upper moves; a cut immediately before the endpoint is excluded.

The two optimizations compared are

$$\max_w \min_r b_{p,r}(w), \qquad \min_r \max_w b_{p,r}(w).$$

The first uses one path at each prime for both rows. The second also allows a different path for each row. They can differ strictly, as the example below shows. After adjoining the previous arithmetic certificate, the resulting common-divisor bounds agree in these tests. This is a finding about the tested certificates, not a general CMF theorem.

We used 89 structured cuts at each of $N = 20, 40, 80, 160, 320, 640$, and 240 seeded random cuts at each of $N = 20, 40, 80, 160, 320, 640, 1280$. The random cuts also vary the progress of the three

lower parameters separately. All arithmetic in these certificates is exact rational arithmetic. The 89-path search found strict gains from prime-dependent selection. Beyond the previous arithmetic normalization, the extra gain factor from row-dependent selection was one at every tested depth. The random family gave the same conclusion about that additional gain.

1.1 A literal prime-dependent order witness

At $N = 80$, let path A first increase each of x_0, x_1, x_2 by $3N$, in that order; it then completes the remaining positive-upper moves, the lower moves, and the negative-upper moves. Path B uses $4N$ instead of $3N$ for that initial block. The cut is taken after the initial block. These are different multiplication orders, since the remaining x_0 moves are performed after some x_1, x_2 moves, with a different split in the two paths. The suffix has all $11N$ negative-upper moves. Their common-row valuation certificates are

	$p = 2$	$p = 59$
A	-19	-3
B	-25	-2

Use A at 2 and B at 59. In the full 89-path search the sets of maximizers at these two primes are disjoint. Thus this benefit is not an artifact of choosing different names for tied optimal paths. At $N = 80$, prime adaptation improves the common factor over the best single fixed path (after including the previous arithmetic bound) by the factor 147649788.

1.2 A strict row-dependent order improvement

At $N = 40, p = 491$, compare the cuts

$$w_A = w_0 + N(3, 3, 3, -3; 0, 0, 0), \qquad w_B = w_0 + N(3, 3, 3, 0; 0, 0, 0).$$

The first moves $3N$ negative-upper steps before completing the remaining positive-upper and lower moves; the second postpones all negative-upper moves until after those moves. Their certificates are

	numerator row	denominator row
A	-1	-2
B	-2	0

In the entire 89-path family, the best common-path minimum is -2. Choosing A for the numerator and B for the denominator gives -1. Thus independent row choices improve the path-only common-divisor certificate by a factor of 491. The earlier arithmetic normalization already certifies valuation -1 at this prime, so this gain does not improve that normalization further.

For a second illustration, at $N = 80, p = 883$, a path with cut

$$w = w_0 + N(6, 6, 6, -9; 2, 1, 3)$$

has row certificates $(-2, -2)$, whereas the $3N$ initial-upper path above has $(-3, 0)$. The first better certifies the numerator row, the second the denominator row. The sets of row maximizers in the 89-path family are disjoint. Nevertheless the common minimum remains -2: row adaptation adds no common factor in this example. The numerator remains the limiting row in the aggregate experiments.

1.3 Finite gains against the previous six-interval bound

Each certificate is combined prime by prime with the earlier normalization. The following are natural logarithms of upper bounds on the *reduced* denominator; the actual reduced denominator is computed independently from the endpoint rational number.

N	Previous bound	89-path bound	Actual $\log q_N$
20	1006.830013	969.294302	966.461089
40	2024.521715	1952.684660	1952.684660
80	3995.558516	3897.972570	3897.279423
160	7952.463445	7845.385158	7843.998864
320	15860.979328	15681.089752	15668.557979
640	31662.180409	31318.172451	31310.353216

These finite gains include primes omitted by the cutoff $p^2 > 50N$; that part contributes no exponential-rate improvement. It must not be extrapolated as a constant saving per depth. The next section isolates a uniform additional divisor among the eligible larger primes.

There is an important distinction: at $N = 320,640$, every newly explained large-prime factor is already exposed by the *single* cut after all positive-upper and lower moves, before the entire negative-upper block. Different prime-dependent paths improve other finite factors, but the new asymptotic divisor below does not require changing the path by prime. It is therefore not evidence that adaptive orders are necessary for the stronger irrationality measure.

2 The uniform arithmetic extension

This is an addendum to the supplied full manuscript *CMF conservation as an arithmetic tool*, saved as `zeta2_complete_proof.tex` in the accompanying bundle. We use its exact native readout, baseline normalization, eight-cell signature table, and local-kernel lemma. Those statements include the initial conditions and all native generator identities. The following argument proves the extension needed here; it does not infer it from the finite path tests.

A uniform periodic extension of the arithmetic divisor. The following statement extends the exact finite-field readout argument. It is not a theorem asserting optimality, or necessity, of different multiplication routes for the two initial rows. Let

$$\mathcal{T} = \{(2,8,1),(3,9,2),(4,10,3),(5,8,3),(6,9,4),(12,9,8)\}.$$

For $N \geq 1$ and $r \in \{0,1,2\}$ define $H^*_{N,r}$ to be the product of primes p with $p^2 > 50N$ for which, with $t = \lfloor N/p \rfloor$, at least one $(k,A,B) \in \mathcal{T}$ satisfies

$$\frac{13N+r}{13t+k} < p < \frac{AN}{At+B}. \tag{L1}$$

Every prime occurs once. The inequalities imply $p < 8N$, so the product is finite. Its $t = 0$ part is precisely the previous divisor $H_{N,r}$. If there are no eligible primes, the product is one.

Theorem 1. *With the same initial rows, trajectory, and factors C_N, G_N as in the preceding argument, for every integer $N \geq 1$ and every $r \in \{0,1,2\}$,*

$$\boxed{\frac{C_N d_{12N} d_{13N+r}}{G_N H^*_{N,r}} RW(N) e_r \in \mathbb{Z}^2.}$$

Proof. The baseline integer normalization with G_N has already been established. It suffices to prove one additional factor p for each prime in (L1). Put $x = N/p = t + f$, where $0 \leq f < 1$. The upper inequality in (L1) says $f > B/A$; the lower says

$$f < \frac{k}{13} - \frac{r}{13p}.$$

Thus f lies in the same one of the six open intervals as in the finite chamber table for the original shell argument. The two intervals with an internal split are divided at $5/11$ and $10/11$, exactly as before.

Write

$$a = \lfloor 8x \rfloor = 8t + a_0, \quad M = \lfloor 11x \rfloor = 11t + M_0, \quad b = 8N - ap, \quad d = 11N - Mp, \quad h = \phi(x) = \phi(f).$$

The equality for h follows directly from the periodic floor formula. For $0 \le j < p$, the carry indicators $e = \mathbf{1}_{j>d}$ and $\delta_u = \mathbf{1}_{(j+uN) \bmod p<b}$ depend only on f and j/p. On their minimal-carry support they therefore have the same fixed values as in the earlier table. The upper digits translate by

$$\lambda_u = \lfloor j/p + ux \rfloor = ut + \lambda_{u,0} \qquad (u = 9, 10, 11).$$

In particular they are still independent of the supporting residue j. The two exceptional fractional split points cannot occur: if $f = 5/11$ or $10/11$, primality forces $p = 11$ and $N = 11t + 5$ or $11t + 10$, contrary to $p^2 > 50N$.

Over $\mathbb{F}_p$ use the same polynomial quotient

$$\mathcal{Q}(X) = \frac{\prod_{u=9,10,11} \binom{X+uN}{b}}{(X^p - X)^{h-1} \prod_{j=0}^{d} (X - j)}.$$

For $h = 0$ the displayed formula means multiplication by $X^p - X$. The carry multiplicities establish polynomial divisibility exactly as before. Its decisive degree parameter is now

$$K = 3a - M + h = 13t + (3a_0 - M_0 + h) = 13t + k.$$

Hence (L1) gives

$$\deg(X^r \mathcal{Q}) = 13N + r - (K-1)p - 1 < p - 1, \qquad \sum_{j \in \mathbb{F}_p} j^r \mathcal{Q}(j) = 0. \tag{L2}$$

For clarity, the local kernel argument also remains valid with these larger upper digits. Set

$$P(V) = \prod_{u=9,10,11} (a+1)^{\delta_u} \binom{\lambda_u + V}{a + \delta_u}, \qquad U(V) = \frac{P(V)}{\prod_{l=0}^{M-e} (V - l)}, \qquad \kappa = (-1)^{d-M+h-1} M! d!.$$

The cutoff implies $p > 50x$; the shell intervals imply $x > 1/8$. Thus p is odd, $a + 1 < p$, and $M < p$, so all factorial denominators and distinct kernel poles used here are legitimate over $\mathbb{F}_p$. The exact separation of factors divisible by p gives, as previously,

$$p^{1-h} R_{\ker}(j + pV) \equiv \kappa \mathcal{Q}(j) U(V) \pmod{p},$$

where $R_{\ker}(s) = F(s)(-1)^{11N}(11N)!/\prod_{l=0}^{11N}(s - l)$ is the rational kernel, not the initial-row matrix. Taking residues supplies the leading coefficients c_l/p^h. Subtracting their partial fractions gives

$$p^{1-h} S_r(j + pV) \equiv \kappa j^r \mathcal{Q}(j) [U(V)]_+ \pmod{p}, \tag{L3}$$

with $[U]_+$ its polynomial part. This subtraction includes the $e = 1$ cases and the noncongruent contributions in the original kernel. Residue classes outside the minimal-carry support give zero. The previously established integer-valued normalization and degree bound below p^2 justify (L3) also at its removable derivative nodes.

Choose the prime-dependent shift

$$\eta = p\left\lceil \frac{2N}{p} \right\rceil = Lp.$$

If $p \le 2N$, then $2N \le \eta < 4N$; if $p > 2N$, then $\eta = p < 8N$. Thus $2N \le \eta \le 9N$ in every case, as required by the exact readout. In the genuinely new periods $t \ge 1$, one has $p < N$ and the stronger bound $\eta < 3N$. All harmonic indices are below $20N < p^2$ (below $14N$ for $t \ge 1$), and the Newton degree plus one is at most $13N + r < p^2$. Also $M + L \le 13x + 1 < p$, so the high-digit harmonic reciprocals are units. The integer factors at the relevant shifted Newton nodes have absolute values below $24N + r < 50N < p^2$.

The harmonic leading residue factors as a constant times the sum in (L2), because the multiples of p up to $j + (l + L)p$ are indexed by $l + L$, independently of $j \in \{0, \dots, p-1\}$. For the polynomial part, (L3) and Lucas's congruence show that every Newton coefficient with index $vp - 1$ has the same factor $\sum_j j^r \mathcal{Q}(j)$ and hence vanishes modulo p. These are the only possibly nonintegral weights in the polynomial functional. Both contributions therefore have valuation at least $h - 1$; the second row has valuation at least h. Finally $v_p(d_{12N} d_{13N+r}) = 2$ and $v_p(G_N) = h$, so both normalized entries acquire the asserted extra p. Multiplying these prime divisibilities proves the statement for all N, including the cases where the eligible product is empty. □

The arithmetic improvement in this theorem comes from the same exact matrix readout and its prime-dependent polynomial analysis. Separate finite experiments may expose these factors through specific multiplication orders; their success alone is not the proof above.

3 The improved rate

Density of the lifted prime intervals. For

$$\mathcal{B} = \{(2,8,1), (3,9,2), (4,10,3), (5,8,3), (6,9,4), (12,9,8)\},$$

write (k, a, b) for an element. The additional intervals, beyond the six original intervals, are

$$\frac{13N + r}{13t + k} < p < \frac{aN}{at + b}, \qquad t = 1, 2, \dots.$$

They are disjoint because their reciprocals lie in the disjoint translated base intervals. If $H^{\mathrm{lift}}_{N,r}$ is the product of eligible primes in these intervals, the prime number theorem gives

$$\lim_{N\to\infty} \frac{\log H^{\mathrm{lift}}_{N,r}}{N} = g := \sum_{(k,a,b)\in\mathcal{B}} \sum_{t=1}^{\infty} \left(\frac{a}{at + b} - \frac{13}{13t + k} \right).$$

To justify the infinite sum, first retain finitely many t. The column shift changes each endpoint only by a bounded amount. The discarded intervals for $t \ge T$ contain only primes $p < N/T$, so their total logarithmic weight is at most $\vartheta(N/T)$; after division by N, the limiting upper bound is $1/T$. The excluded primes $p \le \sqrt{50N}$ contribute $o(N)$. This proves the stated limit without requiring a prime number theorem uniform in t. Equivalently,

$$g = \sum_{(k,a,b)\in\mathcal{B}} \left(\psi(1 + k/13) - \psi(1 + b/a)\right) = 0.10868236921910157039\dots.$$

An exact rational enclosure. Each summand is

$$f_{k,a,b}(t) = \frac{ak - 13b}{(at + b)(13t + k)} > 0.$$

Put $M = 100000$ and $S = 10^{24}$. Integer division gives

$$A := \sum_{(k,a,b)\in\mathcal{B}} \sum_{t=1}^{M} \lfloor S f_{k,a,b}(t) \rfloor = 108681309400407200714213.$$

Since $f(t) = (\beta - \alpha)/((t+\alpha)(t+\beta))$ with $\alpha = b/a < \beta = k/13$, comparison with decreasing reciprocal squares gives

$$\frac{ak - 13b}{a(13M + 13 + k)} \le \sum_{t=M+1}^{\infty} f_{k,a,b}(t) \le \frac{ak - 13b}{13aM}.$$

Consequently

$$\frac{A}{S} + \sum_{\mathcal{B}} \frac{ak - 13b}{a(13M + 13 + k)} \le g \le \frac{A + 6M}{S} + \sum_{\mathcal{B}} \frac{ak - 13b}{13aM}.$$

The accompanying integer-arithmetic script verifies this enclosure:

$$0.1086823692136766 < g < 0.1086823692294671.$$

Using the previously certified carry-density lower bound

$$\gamma \ge \frac{159010385641752609841}{13110552000000000000},$$

set $c_0 = 25 - 233/120 - \gamma_{\text{lower}}$ and let g_- denote the displayed rational lower endpoint. The existing analytic enclosures imply

$$\sigma \le 38.168732909647 + c_0 - g_-, \qquad \tau \ge 24.467842131921 - c_0 + g_- > 0.$$

Exact rational cross-multiplication then gives

$$1 + \frac{38.168732909647 + c_0 - g_-}{24.467842131921 - c_0 + g_-} < 4.589897.$$

This last implication uses the uniform integer normalization furnished by the translated shell argument and the analytic/nonvanishing inputs. It does not follow merely from the finite divisibility checks below.

Why the column shift matters even after translation. At $N = 3000$, $p = 907$, the active lift has $t = 3$, $k = 4$ and $(13t + k)p = 43 \cdot 907 = 39001$. Thus $13N + r < (13t + k)p$ holds only for $r = 0$. For the pair normalized using the original six-shell divisor, its two p-valuations are $(1, 3)$ in column zero, but $(0, 2)$ in columns one and two. The unshifted periodic guess therefore fails, whereas the shifted inequality predicts precisely the observed distinction. The same phenomenon occurs at $N = 6000$, $p = 1013$: $(13 \cdot 5 + 12)1013 = 78001$.

The improved irrationality exponent. Let $H^*_{N,r}$ include all the translated shells proved in the preceding arithmetic lemma. Their additional logarithmic density is

$$g = \sum_{(k,a,b)\in\mathcal{B}} \sum_{t\ge 1} \left(\frac{a}{at+b} - \frac{13}{13t+k} \right), \qquad 0.1086823692136766 < g < 0.1086823692294671,$$

where $\mathcal{B} = \{(2,8,1),(3,9,2),(4,10,3),(5,8,3),(6,9,4),(12,9,8)\}$. The original six shells have density $233/120$. Together with the certified carry-density lower bound, this gives

$$c = 25 - \gamma - \frac{233}{120} - g < 10.821223.$$

The integer pairs supplied by the arithmetic lemma consequently satisfy

$$|Q_N| \le \exp\big((48.989956 + o(1))N\big), \qquad |Q_N\zeta(2) - P_N| \le \exp\big(-(13.646619 - o(1))N\big).$$

Here we used the analytic bounds $a < 38.168733$ and $b < -24.467842$. The previously established bounded-gap nonvanishing property supplies the required nonzero rational comparison among three consecutive indices. The standard irrationality-measure argument therefore implies

$$\boxed{\mu(\zeta(2)) \le 1 + \frac{48.989956}{13.646619} = \frac{62636575}{13646619} = 4.589896955429033\ldots < 4.589897.}$$

All decimal inequalities here have exact rational certificates in the accompanying code. The CMF approximants and their reduced denominators are unchanged: the improvement is in the divisor that the argument certifies.

4 Reproducibility and limits of the search

The bundle includes the previous complete derivation, the present addendum, all new search programs, exact checkpoints, the translated kernel checks, and the rational density certificate. It also records the eight-depth prime-pattern checks for all three output columns. The pattern checks are numerical discovery evidence; the translated finite-field argument above supplies the all-depth statement.

The requested new attachment `zeta2_cmf_matrix_proof (4).tex` was an empty file on disk. The previous nonempty attachment was inspected as background. No claim is made about the contents of the empty file.

The search changes arithmetic descriptions of the same endpoint, so it cannot change the exact reduced approximants or their actual delta values. No new delta search was performed. The previous values were $\delta_{3000} = 0.278673337629$, $\delta_{4000} = 0.279047437501$, $\delta_{6000} = 0.278946586656$, with last-three-depth slope estimate 0.279152068816. These numerical scores are separate from the irrationality-measure conclusion proved using upper bounds.

No extra common-divisor benefit from independent row choice *beyond the previous arithmetic normalization* has been found in the tested families. That remains a target for a richer lattice or residue certificate. The precise next region is: keep the same field, start, and direction; test $N = 640, 1280, 2560$; retain at least $2N$ negative-upper suffix moves; refine cut coordinates in quarter-N increments and allow one or two leading p-adic residue digits. Examine primes outside the periodic divisor with $p^2 > 50N$. At $N = 6000$, remaining eligible residual primes include 1583 in all three columns and 11743 in column 0. A new uniform rule is needed before either can affect the stated measure bound.